

Srujay Reddy Vangoor

AI Software Engineer · LLM Platforms · RAG · Multi-Agent Systems · FastAPI · AWS

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Software Engineer with 6+ years of experience building scalable backend systems, cloud native platforms, and production AI systems. Specialized in designing and deploying LLM based systems including RAG pipelines, multi agent orchestration, evaluation driven workflows, and knowledge graph augmented retrieval. Experienced in building distributed microservices on AWS using Python and modernizing legacy platforms into cloud native architectures.

TECHNICAL SKILLS

AI & LLM Systems: LangChain, LangGraph, CrewAI, RAG Pipelines, Multi-Agent Systems, MCP, Vector Search (FAISS, Pinecone), Evaluation (RAGAS, LangSmith), Prompt Engineering, Hugging Face Transformers, PyTorch, Scikit-learn

Backend & APIs: Python (FastAPI, Django), C# (ASP.NET Core), Node.js (Express), Golang, REST, GraphQL, WebSockets

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, CI/CD, CloudWatch, Grafana, Prometheus

Data Systems: PostgreSQL, MS SQL, DynamoDB, Redis, Kafka, Elasticsearch, Airflow, Spark, Snowflake, Power BI

Frontend & Mobile: React, Next.js, TypeScript, JavaScript, React Native

EXPERIENCE

AI Software Engineer

California Department of Developmental Services (DDS)

Feb 2025 – Present
Sacramento, CA (Hybrid)

- ▶ Built a conversational RAG system for internal report analysis, reducing processing time by 65% and improving grounded response accuracy by 35% using retrieval augmented evaluation pipelines.
- ▶ Engineered a multi-agent orchestration system (CrewAI) with deterministic execution, schema validation, and persistent memory, reducing multi-step failures from 23% to under 7% in production.
- ▶ Improved retrieval quality (hit@1 from 0.35 to 0.63) by designing a knowledge graph augmented retrieval layer with subgraph expansion, hybrid embeddings, and relevance pruning.
- ▶ Designed and deployed AWS cloud native FastAPI microservices supporting 200K+ daily API requests with sub second P99 latency using EC2, ECS, Lambda, S3, IAM, and EventBridge for secure healthcare workflows.
- ▶ Built a React and TypeScript AI workflow intake portal on AWS using Terraform Infrastructure as Code (IaC), reducing submission time from 30 to 10 minutes through automated validation and review.
- ▶ Modernized COBOL, JCL, and Db2 workflows into event-driven Python services aligned with FHIR standards, reducing nightly processing time by 60% and eliminating 200+ manual steps.

Software Engineer

California Department of Conservation (DOC)

Jan 2024 – Feb 2025
Sacramento, CA (Remote)

- ▶ Engineered a seismic modeling service on AWS, integrating Python (OpenQuake) with legacy PHP and Perl systems, delivering ground motion predictions in under 3 seconds per request.
- ▶ Reduced data pipeline latency by 50% by optimizing Kafka-based ingestion of 10+ GB/day from earthquake monitoring stations and migrating analytics from PostgreSQL to Druid for real time analytics.
- ▶ Implemented RBAC, MFA, encryption controls, AWS Secrets Manager, IAM policies, and Azure Entra ID SSO for 60+ monitoring stations, supporting compliance audits and secure access management.
- ▶ Improved geospatial applications performance by 40% by optimizing Leaflet rendering and GeoServer integration for large geospatial datasets, eliminating recurring downtime.

Software Application Developer Intern

Population Research Center (PRC), Sacramento State

Jan 2023 – Jun 2024
Sacramento, CA

- ▶ Built a staff management system using C#, Entity Framework, and SQL Server, optimizing stored procedures and indexing to reduce administrative workload by 40%.
- ▶ Designed and automated health data ETL pipelines using .NET, SQL Server, and PowerShell, migrating Access and Oracle datasets into centralized analytics systems and enabling HIPAA compliant reporting workflows.

- ▶ Developed data entry tools and analytics dashboards for statewide public health studies, increasing data collection efficiency by 60% through automated validation and quality checks.
- ▶ Reduced survey delays by 30% by implementing system upgrades and PowerShell diagnostic utilities to support high volume CATI survey operations.

Software Engineer

Human Sciences Research Group (HSRG), IIIT-H

Jan 2021 – Aug 2022

Hyderabad, India

- ▶ Architected a cloud-native archival and analytics platform for social media and news streams, enabling near real time analysis of 1M+ data points during the Indian Farmers' Protests.
- ▶ Achieved 8% accuracy improvement over baseline by fine-tuning multilingual BERT and IndicBERT using Hugging Face Transformers for sentiment classification and topic modeling.
- ▶ Built 13 secure REST API endpoints using C# and ASP.NET Core for high volume data ingestion and metadata management, with React based dashboards supporting exploratory analysis.

Software Engineer

Change and Continuity in Spiti Valley, ICSSR Sponsored Project

Jul 2018 – Dec 2020

Hyderabad, India

- ▶ Built a GIS platform using Python, Django, PostgreSQL, and PostGIS to map 280+ heritage sites across 25 villages, enabling spatial search, analytics, and interactive visualization.
- ▶ Developed responsive React web and offline capable React Native mobile applications using Leaflet, achieving 60% faster load times in low bandwidth field environments.
- ▶ Automated Python based ETL and spatial processing workflows integrating surveys, census, and archival datasets into a unified geospatial repository, improving metadata accuracy by 40% and enabling near real time updates.

PROJECTS

ContextServer – MCP Tool Infrastructure for AI Agents

- ▶ Built an MCP-based tool server enabling AI agents to inspect and reason over real codebases through filesystem access, Git history, and documentation search.
- ▶ Architected an extensible plugin based tool registry allowing new tools to be added without modifying core routing logic, reducing integration time for new capabilities by 60%.

CA DMV RAG System

- ▶ Built a RAG system using FastAPI, FAISS, and Sentence Transformers over the CA DMV handbook, supporting streaming responses with citations, reranking, multi document retrieval context, and confidence scoring.
- ▶ Deployed with Docker on Render with observability (metrics, distributed tracing), CI/CD via GitHub Actions, and PyTest coverage across ingestion, retrieval, and re-ranking workflows.

VeryFastChat – Anonymous Real-Time Chat Platform

- ▶ Engineered a full stack chat platform using Next.js, FastAPI, and Supabase, supporting instant room creation, shareable invite links, persistent chat history, real time messaging, and host moderation workflows.
- ▶ Secured production deployment on Vercel and Render, startup configuration validation, CORS hardening, rate limiting with Redis support, protected metrics, health monitoring, and GitHub Actions CI and uptime checks.

Buddhira – Personal Knowledge Management App

- ▶ Architected a full stack knowledge management system using Next.js, FastAPI, and Supabase enabling secure note capture, tagging, and search with per-user data isolation.
- ▶ Implemented JWKS based JWT authentication, API rate limiting, and centralized error handling while establishing CI/CD pipelines with automated tests using GitHub Actions.

EDUCATION

Master of Science (MS) in Computer Science

California State University, Sacramento

Aug 2022 – Jan 2025

Sacramento, California

Bachelor of Technology (BTech) in Computer Science

International Institute of Information Technology, Hyderabad (IIIT-H)

Aug 2015 – Jul 2019

Hyderabad, India